

Charting the future of green HRM practices: insights from theories, context, characteristics and methodologies (TCCM) framework and analytical hierarchy process (AHP) analysis

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Abstract

Purpose – The term “green human resource management” (Green HRM) is becoming more prevalent in both academics and business. This study’s main objective is to locate current understanding and trends in Green HRM literature by examining the available Green HRM literature. And prioritize the Green HRM practices in higher education institutions (HEIs). The study also identifies prevailing theories, context, characteristics and methodologies as current research interests to identify potential future research areas in Green HRM.

Design/methodology/approach – The study opted for a multi-method approach, combining theories, context, characteristics and methodologies (TCCM) for literature review and the analytical hierarchy process (AHP) for methodological analysis. The article selection process is illustrated via a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart. The study examines Green HRM literature using the TCCM framework as it aids in identifying commonly used theories, constructs and methods, suggesting future research directions (FRDs). After identifying core Green HRM practices, the study prioritizes them for the HEIs using the AHP.

Findings – The comprehensive analysis of the literature is provided by the TCCM framework, which identifies the most influential theories, contexts, characteristics and methodologies used in Green HRM literature. The most used theory was found to be the ability-motivation-opportunity (AMO) theory, followed by social identity theory. In context, the hospitality industry was the dominating sector/industry of the domain. Under the characteristics, various variables in Green HRM literature were identified and classified as antecedents, mediators, moderators and outcomes. The most used research method was quantitative, and the most opted research design was empirical. FRDs are also suggested analyzing the current trends of the Green HRM field. The finding of AHP indicates that green recruitment and selection holds the highest level of importance among Green HRM in HEIs.

Research limitations/implications – This study is vital for past-future orientation, in which the past is explored in order to build new routes of Green HRM research. It used research publications from the Scopus database, and future studies may use other databases to improve understanding of Green HRM research.

Practical implications – AHP weights and ranks aid decision-makers in assessing Green HRM practice importance, facilitating pairwise comparisons before implementation in the HEIs and other service industries. It guides managers on allocating appropriate importance to each practice for informed decision-making.

Social implications – This study enriches Green HRM literature using the TCCM framework, categorizing literature into four key areas and offering valuable FRDs for future researchers. Its comprehensive review provides crucial insights into the research landscape of Green HRM, aiding in understanding theories, variables, relevant sectors and methodologies. The study offers comprehensive theoretical implications in the realm of Green HRM.

Originality/value – Existing research indicates that no study has employed a multi-method approach combining the TCCM framework and AHP in Green HRM research. The initial TCCM phase conducts a thorough analysis, examining theories, contexts, characteristics and methodology, a unique methodology. It connects TCCM findings to FRDs. In the subsequent phase, AHP prioritizes Green HRM practices for the HEIs based on



identified research gaps from TCCM. This innovative fusion of TCCM and AHP, resulting in FRDs and implications, represents the novelty of this paper.

Keywords Green human resource management, Green HRM, Sustainability, TCCM, Literature review, Analytical hierarchy process, AHP

Paper type Literature review

1. Introduction

Society needs to prepare for the anticipated impacts of climate change, which is expected to lead to significant transformations in socioeconomic systems (Winn *et al.*, 2011). Many scholars have highlighted the link between environmental degradation and climate change, attributing severe weather events, such as floods, droughts, and wildfires to these phenomena, with substantial consequences for lives and livelihoods (Farrukh *et al.*, 2022a, b; Islam *et al.*, 2020). These challenges underscore the need for sustainable measures, including the adoption of Green Human Resource Management (Green HRM) practices within organizations (Chaudhary, 2019).

Green HRM promotes a sustainable culture and encourages pro-environmental practices within organizations and society by reducing energy and natural resource consumption, minimizing travel, creating sustainable workplaces, and lowering carbon emissions (Opatha and Arulrajah, 2014). In addition to lowering environmental impacts, Green HRM promotes employee growth and welfare (Ari *et al.*, 2020) while providing organizations with a competitive advantage by attracting high-quality applicants (Chaudhary, 2019).

In recent years, escalating concerns about environmental degradation and the pressing need for sustainable business practices have heightened interest in Green HRM (Liu *et al.*, 2021). Green HRM has emerged as a crucial component in adopting and maintaining green workplace strategies, enabling organizations to reduce their ecological footprints and foster environmentally responsible activities (Ahmad, 2015). Research in Green HRM has focused on addressing environmental concerns, the effect of corporate activities on the climate, and related economic and social issues (Chowdhury *et al.*, 2023). Despite increased attention from scholars and practitioners, the Green HRM discipline continues to offer development opportunities (Yong *et al.*, 2019). This study aims to contribute to the existing body of research by providing a comprehensive understanding of the Green HRM field. The novelty of this paper lies in its two-stage approach. Phase 1 involves a literature review using the theories, contexts, characteristics, and methodology (TCCM) technique to analyze Green HRM across various dimensions, linking the findings to future research directions (FRDs) and the study's implications. Phase 2 focuses on prioritizing Green HRM practices through the Analytical Hierarchy Process (AHP).

This paper provides a comprehensive review of the current state of Green HRM, drawing insights from recent literature on the intersection of HRM and environmental sustainability. By synthesizing findings from diverse scholarly articles and reputable research sources, this study explores the various dimensions of Green HRM. The selection of research articles was performed using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework. This study examines Green HRM through the lens of the TCCM framework. By examining prevalent theories, techniques, constructs, and research contexts, the TCCM framework identifies frequently utilized ideas, constructs, and methodologies in Green HRM research, offering valuable insights for future research. Therefore, this study aims to address the following research questions:

RQ1. Which theories are used in Green HRM studies?

RQ2. What research contexts have been explored in Green HRM studies?

RQ3. What characteristics are commonly found in Green HRM studies?

RQ4. What methodologies are used in Green HRM studies?

In Phase 2, this study aims to provide useful insights for practitioners, policymakers, and researchers interested in adopting Green HRM practices in their organizations. By prioritizing Green HRM practices through AHP within higher education institutions (HEIs), this study contributes to the ongoing discourse on integrating sustainable practices into HRM. The AHP methodology involves a pairwise comparison based on industry experts' rankings. The approach addresses a gap identified in the TCCM analysis, recognizing that Green HRM remains in its early stages of development within the education sector (Goel *et al.*, 2022). Moreover, HEIs must be competitive, retain talent, enhance their reputation, attract skilled employees, cultivate an eco-friendly culture, improve employee morale, demonstrate social responsibility, ensure sustainability, and contribute to environmental improvement. Adopting green practices can help achieve these objectives, thereby strategizing the development of green educational institutions (Goel *et al.*, 2022). Thus, the AHP is particularly beneficial for prioritizing Green HRM practices, as it offers a structured framework for comparing these practices. By breaking down complex decisions into pairwise comparisons and assigning relative weights to criteria, the AHP facilitates informed and systematic decision-making. It provides a systematic approach to assess and rank options, ensuring the efficient allocation of resources according to their importance and relevance (Khatoon *et al.*, 2022). This study aims to achieve the following two objectives to advance the Green HRM domain.

Objective 1. To conduct a comprehensive literature review using the TCCM framework.

Objective 2. To understand and prioritize core Green HRM practices within HEIs.

2. Review of literature

2.1 Green HRM

Green HRM is a relatively new concept that gained popularity over the past decade as HRM began implementing green management practices to lower its environmental imprint. Since 2013, there has been a significant surge in research on Green HRM practices, recognizing its vital role in promoting environmental sustainability. The origins of Green HRM are often traced to Wehrmeyer's book (1996), which combined and characterized HRM and environmental management as "green." Later, Renwick *et al.* (2013) defined Green HRM as the HRM aspect of environmental management.

Green HRM has emerged as a field due to the broader scope of environmental management research and the growing public expectations for businesses to participate in global environmentalism (Jabbour *et al.*, 2019; Paulet *et al.*, 2021; Pham *et al.*, 2020). Green HRM entails incorporating environmentally friendly practices into conventional HRM tasks, such as recruitment, training, performance evaluation, and employee engagement (Jabbour *et al.*, 2019). A study by Molina-Azorin *et al.* (2021) indicates the differences between HRM and Green HRM practices. Moreover, recent literature suggests that Green HRM evolves when management integrates sustainability into its practices (Chand *et al.*, 2024).

Green HRM, according to Opatha and Arulrajah (2014), refers to the use of organizational policies, practices, and processes that develop green employees for the benefit of the individual, the team, society, the environment, and the organization. Various studies have outlined Green HRM practices, such as green job design, green recruiting and selection, green training and development, green performance management, and green compensation management (Bangwal and Tiwari, 2015; Jabbour *et al.*, 2019; Shahriari *et al.*, 2019). These green practices help achieve environmental goals throughout the HRM process, from recruitment to exit. Green HRM practices benefit organizations and their employees by promoting economic stability, environmental balance, social equity, health, and wellness sustainably (Jiang *et al.*, 2024).

2.2 Green HRM practices

Green job description: Incorporating environmental concerns into employees' job descriptions extends beyond their regular duties to encompass additional environmental responsibilities within their roles (Jabbour *et al.*, 2010). Later, Opatha and Arulrajah (2014) also highlighted this as an HRM function, integrating green policies by including environmental responsibilities in job descriptions.

Green recruitment and selection: This refers to the process of identifying and hiring candidates who not only possess the required qualifications and skills for a job but also demonstrate a commitment to environmental sustainability. This approach ensures that employees are aligned with an organization's eco-conscious values and goals, promoting environmentally responsible practices in the workplace (Firdaus *et al.*, 2015). Recognizing its importance, Pham and Paillé (2020) presented the first literature review on green recruitment and selection, identifying it as a crucial practice for enhancing a job seeker's perception of organizational attractiveness, job pursuit intention, job acceptance, and job recommendation.

Green training and development: It is widely recognized as a critical Green HRM intervention, primarily aimed at raising employee awareness of the environmental impact of their organization's actions and equipping them with essential environmental skills (Renwick *et al.*, 2013). Employee training is essential for the effective implementation of an environmental management system and for fostering an environmentally friendly organizational culture (Shahriari *et al.*, 2019). Recent literature also highlights green training as essential for overcoming individual barriers to environmental protection and fostering acceptance in the workplace, yielding positive outcomes for both organizations and employees (Yafi *et al.*, 2021).

Green performance management: This involves integrating organizational policy with environmental responsibility. The integration of environmental management systems improves the quality and value of environmental performance (Renwick *et al.*, 2013). It also enables firms to achieve more sustainable performance by retaining employees who are knowledgeable about environmental issues. This approach is essential for conserving and preserving the environment and for protecting the community in the long term (Martins *et al.*, 2021).

Green rewards and compensation: This refers to a system of monetary and non-monetary awards for those who contribute to environmental management goals (Jabbour *et al.*, 2010). It also enhances individual environmental performance (Paillé *et al.*, 2023). A recent study by Dash and Das (2024) confirms that green rewards are vital for environmentally conscious organizations to meet their goals of efficient resource utilization and to support the broader mission of environmental protection.

Green employee involvement and empowerment: This is a HRM method that encourages employees to participate in environmental management projects and activities. It empowers employees to make environmentally responsible decisions and fosters a sense of ownership in driving sustainability efforts (Wicaksari *et al.*, 2024). It has emerged as a critical tool for navigating competition and enhancing growth opportunities, serving as a marker of company goodwill (Agarwal, 2017) and also exerting a significant influence on employee retention (Wicaksari *et al.*, 2024).

Besides these core Green HRM practices, various studies have associated other HRM functions with environmental management, including incentive management (Ramus, 2002), discipline management (Opatha and Arulrajah, 2014), employee relations (Ahmad, 2015), green competence building (Pinzone *et al.*, 2016), green self-managed teams (Kundu and Chahar, 2024), corporate cultural management, corporate learning management, and group articulations (Jabbour *et al.*, 2010).

Any critical analysis of the future of Green HRM should be supported by prior research and expertise. Following Table 1 is a summary of previous Green HRM reviews, highlighting their findings and proposed FRDs.

Table 1. Key findings and FRDs from past reviews

Authors	Findings	FRDs
Miah <i>et al.</i> (2024)	This study explores through SLR the influence of both organizational and employee-related factors on the implementation of Green HRM practices within organizations	Suggests future researchers can take circular economy performance area studies into consideration, work with different time frames, and can also study Green HRM for specific industries
Chowdhury <i>et al.</i> (2023)	This paper has explored, characterized, and systematically reviewed multiple theories on HRM, specifically those in the fundamental field of Green HRM research, by adopting SLR approach, critiquing the results found from 127 core research papers	The relationship of Green HRM with other dimensions of sustainability, such as political, institutional, and legal, explores Green HRM from a stakeholder's perspective at societal levels
Bahuguna <i>et al.</i> (2022)	Green HRM has two streams-Conceptual (basic studies) and Empirical (Green HRM and its outcomes)	Explore individual characteristics like values attitude and organizational factors towards environment and check their effect on Green HRM
Paulet <i>et al.</i> (2021)	Surmised important aspects of Green HRM research emerging over the preceding decade. Also conclude with an examination of how Green HRM might grow in the future	Suggests future research from employee (individual), employer, and social perspective
Benevene and Buonomo (2020)	Study showed “training and development”, “performance management and appraisal” as the most frequently used Green HRM practices, while underrepresenting others, such as “Job analysis and description”	Most studies tackle organizational outcomes, failing to address individual, employee-related Green HRM effects; studies tackling organizational outcomes of Green HRM fit in the AMO framework. Explore what kind of requirements pose to the organization, as well as to the daily employee behaviors
Farrukh <i>et al.</i> (2022b)	Identified the most relevant authors, the topics already covered, and new lines for future investigation. Also, analyzing popular issues, research trends, and chronological development Green HRM shows an upward publication trend	Only a handful of studies are there about antecedents of Green HRM, exploring more environmental and social factors. Practitioners need practical knowledge to foster employee behavior in line with the organizations’ mission and vision on sustainable development
Amrutha and Geetha (2020)	Developed a model that conceptually explores the mediational role of “employee green behavior at workplace” in the relationship between Green HRM practices and social sustainability of organizations from employee perspective	Empirical studies can be conducted in future to establish the relationship between Green HRM practices and social line of organizational sustainability, including of possible effects of employee green behavior at workplace, government interventions and trade union
Shahriari <i>et al.</i> (2019)	Summarized Green HRM functions, methodologies used, influential factors identified, and description of gaps in previous studies	Further examine which functions are important in Green HRM and more affected by the green behavior of employees
Pham <i>et al.</i> (2020)	16 detailed recommendations of gaps and research framework with Future Research Directions. Special mentions to technology 4.0, green behavior outside organizations and circular economy	Manufacturing is the most commonly studied industry/economic sector in previous publications, followed by multi-industries and the service sector Further studies need to explore the effects of Green HRM practices on employees’ green attitude/behavior/activities outside of the organization

(continued)

Table 1. Continued

Authors	Findings	FRDs
Ahmad (2015)	Suggested Green initiatives for Human resources like green buildings and paperless offices etc.	–
Renwick <i>et al.</i> (2013)	Classified Environmental management and HRM literature, and identified Green HRM practices under Ability-Motivation-Opportunity model	Testing potential mediators of staff tacit knowledge, employee empowerment, and supportive work cultures
Source(s): Authors' own work		

3. Research methodology

3.1 Phase 1: literature review

The literature review of articles was conducted using the Scopus database, as it is regarded as one of the most comprehensive and widely recognized academic research databases in the world. Scopus has also been utilized in previous systematic review studies. For article selection, a PRISMA statement was followed. To enhance the literature review, the TCCM framework was applied (Paul and Rosado-Serrano, 2019). TCCM helps clarify the origins, evolution, critical areas of emerging research, and potential future directions in any given field. The TCCM framework highlights the primary data sources, participants, contexts, and variables in pertinent research, ultimately providing alternative techniques for analyzing institutional quality and global competitiveness (Paul and Rosado-Serrano, 2019). It also highlights both the theoretical and empirical aspects of a research domain, effectively addressing the limitations of traditional systematic reviews (Chen *et al.*, 2021) and bibliometric analysis (Sharma *et al.*, 2020). This framework serves as a valuable tool for gaining a thorough understanding of a specific research area (Paul and Rosado-Serrano, 2019).

Stage 1. Search strategy

Our research incorporated Green HRM and its synonyms, as well as sustainable HRM and its synonyms, as keywords. This inclusion was made because Green HRM exists both as a distinct field of inquiry and as a subset of sustainable HRM, which serves as an umbrella term for Green HRM (Kramar, 2014).

To fully capture the published literature, the following keywords were used in the search database: “green human resource management” OR “GHRM” OR “green human resource management practices” OR “Green HRM” OR “green human resources” OR “environment* human resource management” OR “environment* HRM” OR “sustainable human resource management” OR “sustainable HRM” OR “sustainable HRM” OR “sustainable human resources” OR “sustainable human resource management practices.”

Stage 2. Screening for inclusion

The search was limited to the language English, and the selected subject areas were Business Management, Accounting, and Social Sciences. To further refine the research, duplicates were removed, and articles were filtered based on a review of their titles and abstracts.

A corpus of 910 papers was initially generated from the database. After applying the inclusion criteria, 576 documents were selected for review. These documents were further screened by thoroughly reviewing their titles and abstracts, resulting in 236 articles. These articles were then carefully examined to distinguish relevant studies from non-relevant studies. Only articles that were relevant to the research objectives and contributed meaningfully to the study were finalized. After reviewing each article in the corpus, a total of 35 articles were selected for the review (see Table 2 and Figure 1).

Stage 3. Data analysis and interpretation

Table 2. Search string and eligibility criteria

Search string	“green human resource management” OR “GHRM” OR “green human resource management practices” OR “Green HRM” OR “green human resources” OR “environment* human resource management” OR environment* HRM” OR “sustainable human resource management” OR “sustainable HRM” OR “sustainable HRM” OR “sustainable human resources” OR “sustainable human resource management practices”
Subject	Business Management, Accounting, and Social Science
Document type	Article, Review
Time frame	2013–June 2024
Eligibility criteria	Focus on Green HRM relevant articles and not its parallel fields such as CSR or sustainable HRM, green supply chain management
Source(s):	Authors’ own work

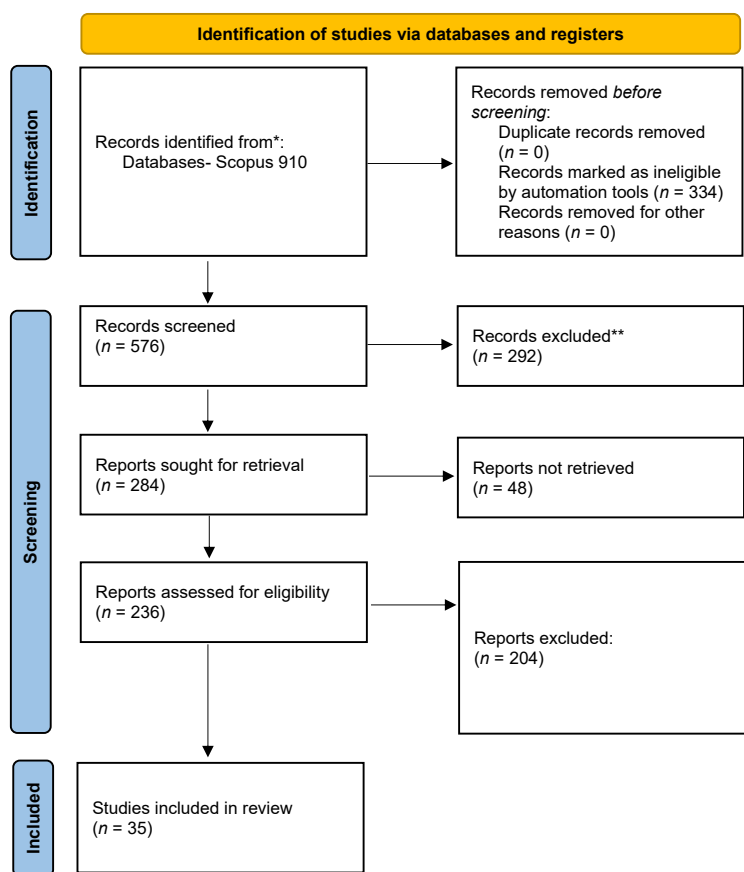


Figure 1. PRISMA flowchart

The final catalog of research publications was thoroughly reviewed to ensure a high-quality literature synthesis. A comprehensive conceptual mapping of the subject matter, aligned with the interpretation of the data within the TCCM framework, was conducted.

3.2 Phase 2: analytical hierarchy process

AHP is a multi-criteria decision-making (MCDM) technique introduced by Saaty (1990) that is used for prioritization and does pairwise comparisons among factors (Toke and Kalpande, 2019).

MCDM is a valuable tool for analyzing decision problems and assessing alternatives according to the values and preferences of the decision-maker. By considering multiple criteria, MCDM techniques systematically evaluate and compare different options, facilitating more informed and rational decision-making (Eldrandaly *et al.*, 2009). This allows decision-makers to identify the most viable solution.

Within the scope of multi-criteria analysis, different methods are used to assign weights to the selected criteria or factors. However, the AHP stands out as the most popular, renowned for its accuracy and prominence in decision analysis, as evidenced by previous studies (Chang *et al.*, 2008; Khatoon *et al.*, 2022).

In this study, the AHP is used to create a matrix for comparing the importance of Green HRM practices. AHP assigns weights, categories, and values based on specific circumstances through the recognition of obstacles.

To prioritize the Green HRM practices, nine experts were interviewed and asked to complete the matrix. All nine experts were permanent faculty with at least five years of industry experience. Each interview lasted around 30–45 min, during which all practices were explained, and experts filled in the responses based on their preferences. The interviews were conducted primarily in English, with some instances of Hindi (local language).

All practices were compared based on their importance, ranging from “equally important” to “extremely important.” The upper right side of the matrix was filled using the scale presented in Table 3, adopted from Saaty (1990).

3.2.1 Steps to conduct AHP analysis.

Step 1. Develop a hierarchical structure with the goal at the top level, attribute or criteria at the second level, and alternatives at the third level, as depicted in Figure 2.

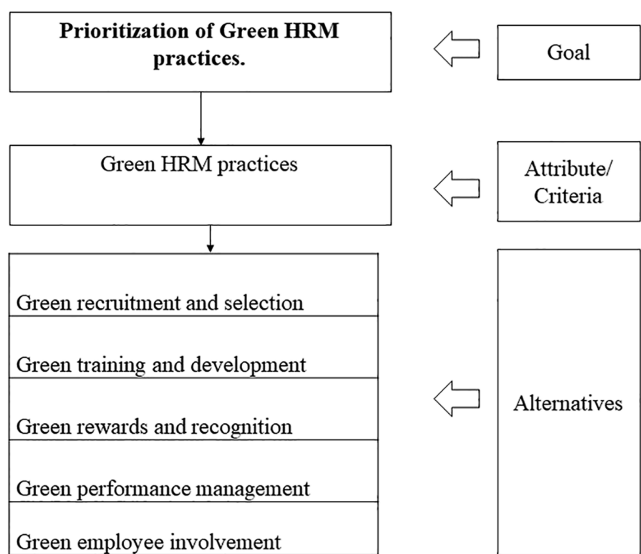
Step 2. Create a pairwise comparison matrix to determine the relative importance of different attributes or criteria with respect to the goal. The matrix is filled using the scale of relative importance present in Table 3.

Step 3. Average the responses from experts and then normalize the pairwise comparison matrix by dividing all the elements in each column by the sum of that column.

Table 3. Scale of relative importance

Intensity of importance on an absolute scale	Definition	Explanation
1	Equal importance	Both practices contribute equally towards Green HRM
3	Moderate importance	Experience and judgment moderately in favor one factor over another
5	Essential or strong importance	Experience and judgment strongly favor one factor over another
7	Very strong importance	Practice is strongly favored
9	Extreme importance	The evidence favoring one practice over another is of the highest possible order of affirmation
2, 4, 6, 8	Intermediate values between the two adjacent judgments	When compromise is needed

Source(s): Saaty (1990)



Source(s): Authors' own work

Figure 2. AHP – hierarchical structure

Step 4. Calculate the criteria weights by averaging all the elements in each row.

Step 5. Check the consistency by calculating the eigenvalue to determine the consistency index (CI). The consistency is finally checked using the consistency ratio (CR), obtained from Equations (1 and 2). If the consistency ratio of a decision matrix is greater than 0.1, it is considered inconsistent and should be recalculated with expert guidance.

$$CI = (\lambda_{\max} - n) / (n - 1) \quad \text{Equation(1)}$$

$$CR = CI / RI \quad \text{Equation(2)}$$

where $\lambda_{\max}$ is the largest eigenvalue of the matrix, n is the dimension of the matrix, and R is the random index, which is based on the values in Table 4.

Step 6. Ranks are assigned based on the criteria weights.

4. Results and discussions

4.1 Theories

4.1.1 Green HRM within the framework of ability-motivation-opportunity theory. Initially, Green HRM practices were defined in the context of ability-motivation-opportunity (AMO)

Table 4. Random index table

Matrix size	1	2	3	4	5	6	7	8	9	10
Random consistency index	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

Source(s): Saaty (1990)

theory by [Renwick et al. \(2013\)](#). Since then, this theory has been expanded and applied in various recent studies, such as those by [Amrutha and Geetha \(2020\)](#), [Benevene and Buonomo \(2020\)](#), [Fawehinmi et al. \(2024\)](#), [Hooi et al. \(2022\)](#), [Iftikar et al. \(2022\)](#), and [Zhang et al. \(2024\)](#). Overall, studies examining organizational outcomes of Green HRM practices, whether assessed as a general score or through specific dimensions, consistently align with the AMO framework ([Benevene and Buonomo, 2020](#)). A recent study also utilizes AMO theory to investigate the impact of Green HRM practices on environmental performance ([Al-Alawneh et al., 2024](#)).

AMO theory is also the primary theory for comprehending the impact of HRM functions on organizational performance ([Table 5](#)). The greening of HRM and its environmental consequences can be better understood through this theory. AMO classifies Green HRM practices based on three major factors: ability, motivation, and opportunity ([Amrutha and Geetha, 2020](#); [Benevene and Buonomo, 2020](#)). Employees' abilities are based on practices, such as recruiting and selection, training, and development programs that guarantee they have the knowledge and skills required to accomplish specific duties. Motivation comprises performance appraisal and both financial and non-financial incentives designed to boost employees to attain their goals. Finally, opportunity encompasses practices, such as participation, information sharing, and methods for increasing independence ([Marin-Garcia et al., 2016](#)).

4.2 Context of green HRM

4.2.1 Industries studied in the context of green HRM. At the beginning of the millennium, discussions on environmental management focused on manufacturing firms. However, activities in the services industry also significantly contribute to climate change, species extinction, and various forms of pollution, including sound and light pollution, as well as the loss of natural resources ([Tanova and Bayighomog, 2022](#)). In recent years, environmental issues in the services sector have gained prominence, with many service firms adopting environmental management systems aimed at pollution prevention. Service firms have increasingly recognized the importance of safeguarding the environment ([Ari et al., 2020](#)), preserving water and energy, and addressing their impact on local and global habitats to maintain a sustainable competitive advantage ([Tanova and Bayighomog, 2022](#)). In the service sector, higher education organizations have, in the past few years, taken significant steps towards incorporating environmental management and green practices into their operations. As teaching and research organizations, higher education organizations are expected to play a significant role in adopting approaches and alternates to address existing environmental issues ([Aboramadan, 2022](#)).

A considerable body of research on Green HRM has focused on the hospitality industry ([Haldorai et al., 2022](#); [Islam et al., 2021](#); [Kim et al., 2019](#); [Muisyo et al., 2022b](#); [Raza and Khan, 2022](#); [Yeşiltaş et al., 2022](#)). However, studies on Green HRM within HEIs remain limited ([Aboramadan, 2022](#); [Akhtar et al., 2022](#); [Chaudhary, 2019](#)). As illustrated in [Figure 3](#), research on Green HRM has been more prevalent in the service sector, with a notable gap in the education industry. It is crucial to highlight that HEIs have both the capability and responsibility to teach society's stakeholders about environmental greening, thereby fostering an understanding of the importance of adopting a pro-environmental position for present and future generations ([Goel et al., 2022](#)).

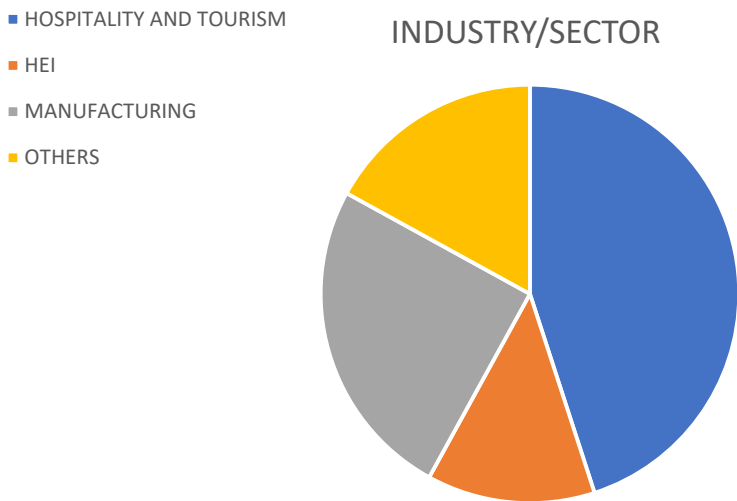
4.2.2 Participants in green HRM studies. Studies on Green HRM primarily focus on employees as research participants, highlighting an emphasis on understanding how employees engage with and respond to Green HRM practices ([Benevene and Buonomo, 2020](#)). Additionally, some studies include both employees and managers, reflecting a broader interest in the hierarchical dynamics of Green HRM implementation. However, research involving other stakeholders, such as prospective employees (e.g. students), is notably limited, suggesting that these groups are underexplored in the current literature. [Table 6](#) highlights this gap.

Table 5. Theories

Theory	Rationale	References
AMO theory	Performance is a result of the interaction of employees' capacity to perform (ability), willingness to perform (motivation), and opportunity to perform through participation (opportunity)	Al-Alawneh et al. (2024) , Ahmad (2015) , Ahmed et al. (2021) , Al-Swidi et al. (2021) , Sabokro et al. (2021) , Yeşiltaş et al. (2022)
Social identity theory	It posits that subjects develop their sense of who they are on the basis of, their belongingness to some specific groups, such as age group and gender organizational membership	Ahmad (2015) , Ali et al. (2022) , Chreif and Farmanesh (2022) , Liu et al. (2021) , Raza and Khan (2022) , Umrani et al. (2022)
Resource-based view theory	The possession of strategic resources provides an organization with a golden opportunity to develop competitive advantages over its rivals. These competitive advantages in turn can help the organization enjoy strong profits	Haldorai et al. (2022) , Ojo (2022) , Yong et al. (2019)
Social exchange theory	People make decisions by consciously or unconsciously measuring the costs and rewards of a relationship or action, ultimately seeking to maximize their reward	Aboramadan (2022) , Ahmad and Umrani (2019) , Yeşiltaş et al. (2022)
Self-determination theory	It pertains to the motivation behind people's choices in the absence of external influences and distractions	Ojo (2022)
Supply-value fit theory	Postulates a joint influence of personal and environmental characteristics on outcomes. Relevant personal characteristics include values and abilities, while the environmental characteristics are supplies and demands	Islam et al. (2020) , Liu et al. (2021)
Person-organization fit theory	This study proposes that the correspondence between the person and the organizational factors (manifested by HRM green practices) may have an impact on individual perceptions and attitudes	Ababneh (2021) , Muisyo et al. (2022b)
Social learning theory	The employees learn by observing their leaders' behavior at the workplace. This theory sheds light on two learning processes: observational learning and reinforcement learning	Ahmad (2015) , Islam et al. (2021) , Tuan (2022)
Theory of planned behavior	The Theory of planned behavior is a psychological theory that links beliefs to behavior. The theory maintains that three core components, namely, attitude, subjective norms, and perceived behavioral control, together shape an individual's behavioral intentions	Omarova and Jo (2022) , Rubel et al. (2021)

Source(s): Authors' own work

Organizations that implement Green HRM are often perceived as more attractive employers by prospective candidates. [Chaudhary \(2019\)](#) emphasizes Green HRM's potential to position organizations as talent magnets, making them appealing to top-quality applicants. Further studies in this area could encourage HRM practitioners to integrate Green HRM into their policies, practices, and employee initiatives, thereby enhancing their ability to attract and



Source(s): Authors' own work

Figure 3. Green HRM studies sector-wise chart

Table 6. Research participants

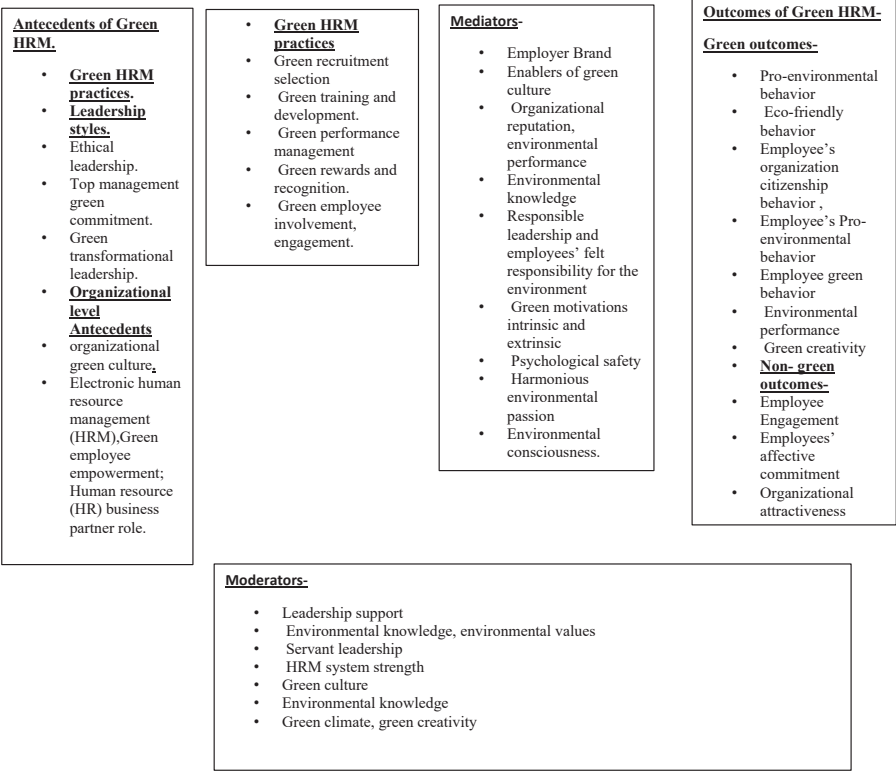
Research participants	References
Employees	Al-Alawneh <i>et al.</i> (2024), Haldorai <i>et al.</i> (2022), Hosseini <i>et al.</i> (2022), Islam <i>et al.</i> (2021), Muisyo <i>et al.</i> (2022a, b), Raza and Khan (2022)
Employees and managers	Ali <i>et al.</i> (2022), Parida <i>et al.</i> (2021), Sabokro <i>et al.</i> (2021)
Prospective employees (students)	Akhtar <i>et al.</i> (2022), Chaudhary (2019)
Source(s): Authors' own work	

retain skilled talent. While most studies focus on organizational-level analysis, fewer studies explore the individual-level impacts of Green HRM (Molina-Azorin *et al.*, 2021), suggesting a promising avenue for future research.

4.3 Characteristic

Research in the Green HRM domain can be characterized by the various variables identified as antecedents, mediators, moderators, and outcomes in the literature. As shown in Figure 4, these variables are categorized into broad domains: Green HRM antecedents, mediators, moderators, and outcomes. The antecedents are further divided into three clusters: Green HRM practices, leadership factors, and organizational factors. The outcomes are divided into two categories: green outcomes and non-green outcomes. Mediators are classified into individual-level and organizational-level factors.

4.3.1 Antecedents of green HRM. The antecedents of Green HRM reviewed in this study are limited, as most studies treat Green HRM as the independent variable, with its practices such as green recruitment selection, green training and development, green performance management, green rewards and recognition, and green employee involvement and engagement acting as drivers for Green HRM. The other two primary clusters of Green HRM antecedents are



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Figure 4. Variables of Green HRM

leadership factors, which describe how top executives' leadership style towards environmental sustainability can influence how they guide their organization's green agenda. This includes leadership styles such as ethical leadership (Ahmad and Umrani, 2019; Chreif and Farmanesh, 2022). Top management's green commitment (Haldorai et al., 2022) and green transformational leadership (Farrukh et al., 2022a) have been identified as antecedents to Green HRM. Another cluster consists of organizational factors; for instance, Yeşiltaş et al. (2022) studied organizational green culture as a driver of Green HRM. Additional factors, such as electronic HRM, green employee empowerment, and human resource business partners, have also been studied as antecedents of green HRM in both the manufacturing and service sectors (Yusliza et al., 2017).

4.3.2 Outcomes of green HRM. The outcomes of Green HRM have been widely studied at both individual and organizational levels, although fewer studies have focused on distinguishing between green and non-green outcomes. In Green HRM literature, outcome variables are typically divided into two clusters: green outcomes and non-green outcomes. Among these outcomes, the majority of studies have examined the influence of Green HRM on various types of employee green behavior, such as pro-environmental behavior (Akhtar et al., 2022), eco-friendly behavior (Raza and Khan, 2022), employee organization citizenship behavior (Tuan, 2022), employee's pro-environmental behavior (Farrukh et al., 2022a), employee green behavior (Chreif and Farmanesh, 2022; Yeşiltaş et al., 2022), environmental

performance (Al-Alawneh *et al.*, 2024; Raza and Khan, 2022), and green creativity (Muisyo *et al.*, 2022b).

Among the non-green outcomes, employee-level factors include employee engagement (Hosseini *et al.*, 2022) and employees' affective commitment (Raza and Khan, 2022). Organizational-level outcomes include factors such as organizational attractiveness (Umrani *et al.*, 2022).

4.3.3 Mediators and moderators. In Green HRM literature, both individual and organizational level variables serve as intervening mechanisms between Green HRM and its outcomes. At the organizational level, studies have identified mediating variables such as employer brand (Hosseini *et al.*, 2022), enablers of green culture (Muisyo *et al.*, 2022a), organizational reputation, and environmental performance (Umrani *et al.*, 2022). On the individual level, some studies have illustrated variables such as environmental knowledge (Akhtar *et al.*, 2022), responsible leadership, employees' felt responsibility for the environment (Tuan, 2022), intrinsic and extrinsic green motivations (Ahmed *et al.*, 2021), psychological safety (Ahmad and Umrani, 2019), harmonious environmental passion (Chreif and Farmanesh, 2022), and environmental consciousness (Yeşiltaş *et al.*, 2022). There are few studies in which Green HRM has been studied as a mediating variable (Ahmad and Umrani, 2019; Chreif and Farmanesh, 2022; Farrukh *et al.*, 2022a; Haldorai *et al.*, 2022; Yeşiltaş *et al.*, 2022).

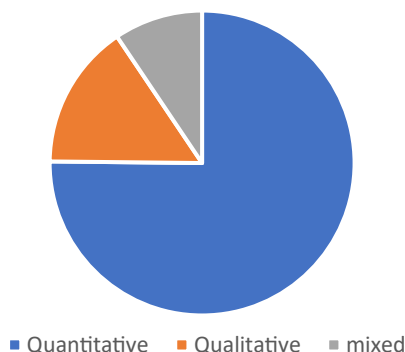
The boundary conditions of the link between Green HRM and its network of outcomes are identified as moderating variables. Various moderators in Green HRM literature include leadership support (Akhtar *et al.*, 2022), environmental knowledge, environmental values (Raza and Khan, 2022), servant leadership (Muisyo *et al.*, 2022b), HRM system strength (Tuan, 2022), green culture (Umrani *et al.*, 2022), environmental knowledge (Farrukh *et al.*, 2022a), green climate, and green creativity (Chreif and Farmanesh, 2022).

4.4 Methodologies (significant methodologies in green HRM research)

Literature provides evidence about the research methodologies (Figure 5), used in past studies. Most scholars have used quantitative methods, followed by qualitative and mixed methods. In research designs (Figure 6), empirical research is dominant, with the survey method being the most commonly used approach.

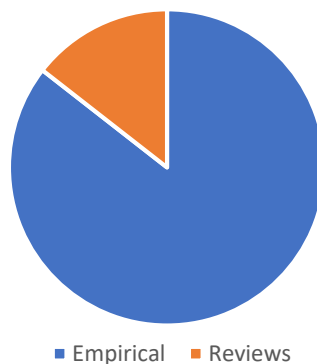
4.5 Analytical hierarchy process (AHP) analysis

Addressing the industry gap found through TCCM, the AHP was conducted in HEIs. The AHP analysis reveals the ranking of Green HRM practices within HEIs: This study examines the



Source(s): Authors' own work

Figure 5. Research methods



Source(s): Authors' own work

Figure 6. Research designs

relative importance of various dimensions of Green HRM practices by assessing their hierarchical significance. Five core green HRM practices, green recruitment and selection, green training and development, green rewards and recognition, green performance management, and green employee involvement were ranked based on pairwise comparisons conducted by industry experts. These dimensions form a matrix to assess their level of importance, and weights were calculated based on the assigned matrix using the normalization technique to derive the relative importance of each practice. Afterward, ranks were allotted based on the criteria weights. The results of the AHP significantly highlight the importance of green recruitment and selection, which was ranked highest with a criteria weight of 0.464665, followed by green training and development at 0.240385 and green rewards and recognition at 0.175272. Green performance management and green employee involvement ranked lower with weights of 0.066519 and 0.053159, respectively. The consistency ratio of 0.067944 indicates consistency within the average matrix, as it is below the threshold of 0.1 suggested by Saaty (1990), thereby validating the results (Table 7).

5. Implications

5.1 Theoretical implications

This study makes several notable theoretical contributions through its literature review, TCCM framework analysis, and AHP analysis. It synthesizes the findings and FRDs from previous literature reviews, consolidating the green HRM studies under the TCCM framework and deriving FRDs from these studies. The application of the TCCM framework strengthens the field's theoretical foundation, and identifying essential ideas, contexts, and features enhances the understanding of the theories, variables, and methodologies currently used in Green HRM literature. These findings enable researchers and practitioners to establish a more solid theoretical framework for implementing long-term HRM initiatives. The FRDs analyzed from the TCCM framework will also guide future researchers. The review highlights the significance of Green HRM in the HEIs within the service sector. An in-depth exploration of Green HRM practices can encourage academia to better understand and focus on implementing green practices within their organizations. Regarding conceptual contributions, this study provides a thorough list of antecedents, outcomes, mediators, and moderators studied in Green HRM literature to date.

This study also contributes by prioritizing Green HRM practices for HEIs. It enriches the Green HRM field by using an MCDM technique such as AHP, which enables the pairwise comparison of practices simultaneously. This methodological contribution paves the way for

Table 7. AHP results

Green HRM practices	Green recruitment and selection	Green training and development	Green rewards and recognition	Green performance management	Green employee engagement	Criteria weights	Ranks
Green recruitment and selection	1	3	4	6	5	0.464665	1
Green training and development	0.333333	1	1	5	7	0.240385	2
Green rewards and recognition	0.25	1	1	4	3	0.175272	3
Green performance management	0.166667	0.2	0.25	1	2	0.066519	4
Green employee involvement	0.2	0.142857	0.333333	0.5	1	0.053159	5
Source(s): Authors' own work							

the broader use of MCDM techniques in related domains, allowing for more thorough analyses.

5.2 Practical implications

This study has practical implications for promoting environmental sustainability by highlighting the context of green HRM. It catalyzes integrating environmental sustainability into organizations. Businesses that employ Green HRM practices assist society by reducing their environmental footprint, conserving resources, and mitigating the consequences of climate change. From an industry perspective, this study focuses on the dominant sectors contributing to Green HRM research, providing insights into where this discipline is most active (see Figure 3). Understanding the industry-wise distribution of research can help identify industrial patterns and potential areas for implementation.

The ranking of Green HRM practices for the HEIs holds significant implications for policymakers, as HEIs are where individuals first develop awareness of environmental protection (Goel et al., 2022). Firstly, it provides insights into which practices should be prioritized and allocated resources effectively. For instance, focusing on green recruitment and selection suggests the need to invest in sustainable hiring practices. Secondly, policymakers can tailor policies and regulations to incentivize the adoption of higher-ranked practices, promoting environmental sustainability within the HEIs and the broader service sector. Lastly, understanding the relative importance of each practice enables policymakers to develop targeted strategies for fostering a green organizational culture and achieving long-term sustainability goals.

In essence, this study not only consolidates the existing body of knowledge in Green HRM but also paves the way for future research and collaboration. It serves as a valuable resource for academics, practitioners, and policymakers who want to make informed decisions, explore new research avenues, and contribute to the rapidly expanding field of Green HRM.

6. Limitations

In this study, it is important to acknowledge certain limitations that emerged during the research process. Despite providing valuable insights, the database and screening process may likely have overlooked some crucial studies. As we only consulted the Scopus database, some pertinent articles may not have been included in our corpus. Only articles from journals were considered suitable for inclusion because they met all of the required standards. As a result, information from other sources, such as books, book chapters, conference proceedings, and errata, was not included in this study. Additional empirical study is required to investigate these concerns more in-depth, enriching the findings further.

7. Future research directions

The FRDs for Green HRM, based on the TCCM framework, span four key dimensions:

- (1) *Theoretical diversification*: Future research could expand beyond the dominant AMO theory by incorporating frameworks such as self-determination theory, which emphasizes intrinsic motivation, social learning theory for observational learning, and the theory of planned behavior, which links attitudes to actions. Additionally, theories such as supply-value fit and person-organization fit could be explored to examine how aligning individual values with organizational goals enhances green behaviors (Liu et al., 2021; Ababneh, 2021).
- (2)
 - *Contextual expansion*: Most current research is sector-specific, particularly focused on industries such as hospitality (Pham et al., 2020). Future studies should consider

multi-sectoral research, exploring both service and manufacturing industries or under-researched sectors such as HEIs.

- *Diverse research participants:* Green HRM studies predominantly focus on employees, leaving a gap in understanding potential employees' perspectives. Future research could investigate how potential employees, such as students, view Green HRM to understand the next generation's expectations.
- (3) *Various characteristics:* HRM is viewed as a technique by managers and management experts to ensure that employees, as individuals or groups, develop attitudes and behaviors that align with organizational strategies. As such, different types of employee green behavior, such as pro-environmental behavior ([Akhtar et al., 2022](#)), eco-friendly behavior ([Raza and Khan, 2022](#)), employee organization citizenship behavior ([Tuan, 2022](#)), employee pro-environmental behavior ([Farrukh et al., 2022a](#)), employee green behavior ([Chreif and Farmanesh, 2022](#); [Yeşiltaş et al., 2022](#)), are widely emphasized outcomes. Therefore, future researchers can expand the green HRM domain by exploring more non-behavioral variables.
- (4) *Methodological approaches:* Most studies in Green HRM have been quantitative, with few incorporating mixed-method approaches. Using mixed methods could provide richer insights and deeper understanding.

Moreover, as suggested by [Bahuguna et al. \(2022\)](#), [Benevene and Buonomo \(2020\)](#), and [Paulet et al. \(2021\)](#), more individual factors must be explored in Green HRM literature. Future researchers could explore Green HRM from stakeholders and society ([Amrutha and Geetha, 2020](#); [Chowdhury et al., 2023](#)). Studies by [Amrutha and Geetha \(2020\)](#), [Farrukh et al. \(2022a\)](#), [Pham et al. \(2020\)](#), and [Shahriari et al. \(2019\)](#) also suggest the need for more work on employee green behavior. To successfully implement an environmental sustainability strategy, organizations must foster green attitudes and behaviors within their workforce, ensuring alignment with the organization's green goals ([Chaudhary, 2020](#)).

8. Conclusions

This study provides useful insights for practitioners, policymakers, and researchers interested in adopting Green HRM practices in their organizations. It also contributes to the ongoing discussion about integrating sustainable practices into HRM by conducting a comprehensive review of contemporary green HRM literature. Ultimately, this study supports a more environmentally conscious approach to managing human capital, as reflected in the findings and FRDs of previous Green HRM reviews presented in [Table 1](#).

The comprehensive analysis of the literature is guided by the TCCM framework, which identifies the most influential theories, contexts, characteristics, and methodologies used in Green HRM research. TCCM helps clarify critical areas of emerging research and potential future directions in the Green HRM field. TCCM framework highlights the primary data sources, participants, contexts, and variables in pertinent research, highlighting alternative techniques for analyzing institutional quality and global competitiveness. All the research questions were addressed using TCCM. In response to [RQ 1](#), the most commonly used theory was AMO theory, followed by social identity theory. In response to [RQ 2](#) (Context), most Green HRM research focused on employees as research participants, with the hospitality industry being the dominant sector. For [RQ 3](#) (Characteristics), various variables in Green HRM literature were identified and classified as antecedents, mediators, moderators, and outcomes. Lastly, in answer to [RQ 4](#), this study provides evidence about the methodologies used in previous studies. The majority of scholars employed quantitative methods, followed by qualitative and mixed methods. In research designs, empirical research is prevalent, with the survey approach being the most commonly used. Despite the growing interest in Green HRM within the research community, this study identified significant opportunities for further

investigation. Based on the answers to the research questions, this study found gaps in green HRM literature and provided recommendations for future research.

This TCCM framework-based review presents a general overview of the Green HRM domain, aiming to offer decision-makers a comprehensive understanding of the field. It enables them to make informed decisions based on an accurate and scientific understanding of the current Green HRM landscape. Recognizing the significance of Green HRM practices in HEIs, this study prioritized core practices, with green recruitment and selection identified as most critical, highlighting their key role. Conversely, green employee involvement ranked lowest among the practices assessed. This prioritization underscores the need for sustainable recruitment strategies to foster environmental responsibility within HEIs.

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